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Survey Experiments in Sociology

Ariela Schachter[1] and Katherine Weisshaar[2]

[1]Department of Sociology, Washington University in St. Louis, St. Louis, Missouri, USA; email: ariela@wustl.edu

[2]Department of Sociology, Northwestern University, Evanston, Illinois, USA

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Keywords

survey experiments, research methods, causal inference, experimental design, surveys, experiments

Abstract

Survey experiments are an underutilized but powerful tool for sociologists interested in studying causal research questions. Survey experiments can yield insights into the breadth of causal relationships, by studying treatment effects in population samples or across subgroups, and can yield a deeper understanding of causal processes that are not readily observed with other social science methodologies. In this article, we begin by considering the conditions under which survey experiments are a uniquely useful method and highlight emblematic examples of recent sociological research. We then discuss some of the challenges and limitations of survey experiments as a research method before offering a brief practical guide to sociologists interested in conducting survey experiments. We conclude with reflections on the future of survey experimental research in sociology.

INTRODUCTION

Survey experiment: an experiment embedded within a survey

Random assignment: a central aspect of experimental design whereby participants are assigned at random to treatment conditions

How do White individuals racially classify others in the United States? Why do parents who temporarily take time out of the labor force to care for young children struggle to regain employment? What are the penalties for reporting sexual harassment? What factors affect beliefs surrounding the denial of healthcare to transgender individuals? How does exposure to populism affect out-group antipathy? These questions have been featured in recent sociological publications (Bonikowski & Zhang 2023, Doan & Grace 2022, Hart 2019, Rivera & Tilcsik 2023, Schachter et al. 2021, Weisshaar 2018), which span multiple distinct subfields in sociology but share the common approach of using survey experiments.

Surveys consist of "(relatively) systematic, (mostly) standardized approaches to collecting information. . .through questioning systematically identified samples" (Marsden & Wright 2010, p. 3), while an experiment is "a study where an intervention [. . .] provides the primary mechanism by which one attempts to make a causal claim" (Druckman 2022, p. 2). Taken together, survey experiments allow social scientists to ask a causal question—a question that "invites a comparison between two states of the world: one in which some sort of intervention is administered (a treated state, i.e., exposing a subject to a stimulus) and another in which it is not (an untreated state)" (Druckman et al. 2011, p. 16). A fundamental challenge is that it is not possible to observe both the treated and untreated states simultaneously within the same person/entity (Druckman et al. 2011, p. 16; see also Holland 1986). Observational research tackles this challenge by trying to account for co-occurring factors that might be related to both the treatment and the outcome. Yet, even with sophisticated statistical techniques, observational research cannot address all possible confounding factors and may even introduce bias in trying to do so, raising many questions about causality (Elwert & Winship 2014).

Experiments take a different approach to this challenge. Experimental research involves random assignment of participants into experimental conditions. Since the participants randomly assigned to the experimental groups should be otherwise equivalent prior to the intervention, random assignment "provides a basis for assuming that the control group behaves as the treatment group would have behaved had it not received the treatment (and vice versa)" (Druckman et al. 2011, p. 17). By comparing the outcome of the treated group to that of the control group,[1] researchers can reasonably claim that the intervention (or experimental manipulation) caused a difference in outcomes. Experimental manipulations can range from changing a single word/phrase (Schachter 2021) to multiple sentences or paragraphs (Warren & Valentino 2023) to images, videos, and other stimuli (Owens 2022).

Different types of experiments are designed to address different types of causal questions (Jackson & Cox 2013, Mize & Manago 2022). Lab experiments are particularly well-suited for data collection requiring in-person measurement and/or interaction (Bonacich & Light 1978), while field experiments offer causal evidence of real-world behaviors or outcomes (e.g., Baldassarri & Abascal 2017). Natural experiments also offer causal leverage and are similar to field experiments in that they capture real-world behaviors and outcomes; however, treatment assignment occurs through some kind of outside force rather than intentional randomization by researchers (Dunning 2012).

In contrast, survey experiments typically consider causal effects in diverse and (relatively) large samples, and for questions that may be difficult or impossible to (ethically) study in a real-world setting. By leveraging both recent advances in experimental design (Mutz 2021) and increased

[1] As we detail below, "control group" here means comparison group, an alternative to the treatment condition. A control group does not have to be a neutral or true null; for example, in studies of racial discrimination, White is often the control condition.

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access to high-quality, Internet-based survey samples (Mullinix et al. 2015), survey experiments can both yield insights into the breadth of causal relationships—studying causal effects in a population sample, or illustrating heterogeneity in such effects across subgroups within the population—and provide deeper understanding of causal processes that are not readily observed with other social science methodologies. These combined strengths make survey experiments an exciting and, we argue, underutilized, option in sociologists' methodological toolkits.

We first examine the characteristics of sociological studies where survey experiments are particularly useful and provide examples of representative research. Our goal is not to provide a comprehensive overview of all survey experiment research in sociology but to highlight studies that illustrate the strengths of survey experiments. Next, we consider some of the challenges and limitations that survey experiments present, before offering a practical guide to conducting a survey experiment. We conclude with thoughts on the future of survey experiments in sociology.

WHEN ARE SURVEY EXPERIMENTS UNIQUELY USEFUL?

We consider survey experiments to be a useful method for research tackling causal questions when

1. The outcome can be well represented by a survey item or is not otherwise easily observable.
2. The experimental manipulations are impossible or unethical outside of a hypothetical context.
3. The counterfactual or counterfactuals include uncommon combinations and/or require an all-else-equal comparison.
4. The hypothesis or hypotheses require comparison across multiple subgroups and/or a large sample for sufficient statistical power.

Most successful studies using survey experiments can be described as having two or more of these characteristics. Below, we expand on each characteristic and offer emblematic examples of recent sociological research of each in turn.

Outcome Can Be Best or Only Observed as a Survey Item

Not surprisingly, survey experiments are well-suited to questions for which a survey item is a good representation of the outcome. Attitudes or perceptions tend to work well as outcomes in survey experiments (Marsden & Wright 2010, Schaeffer & Dykema 2020). But survey experiments are also reasonable when it would be unethical to promote or induce an outcome outside of a hypothetical context, such as prejudicial behavior, or when a behavior is not easily observable—including when social desirability bias leads respondents to mask their true beliefs to a researcher (e.g., Horiuchi et al. 2022). Finally, survey experiments are useful when the goal is to demonstrate how measurement of the outcome itself influences the results. We next address each of these situations related to the outcome in turn.

First, survey experiments are ideal for addressing questions about the causes of differences in attitudes, perceptions, or beliefs—outcomes often observed as survey items (Marsden & Wright 2010, Schaeffer & Dykema 2020). Survey experiments offer the strengths of a population-based survey—namely, representative data and/or the ability to test for differences among subgroups (see more below)—with the strengths of an experiment, i.e., causality. Sociological scholarship interested in support for social policies, or the effects of policies on attitudes or beliefs, illustrates these characteristics well. For example, Philipp and colleagues (2023) demonstrate how information about parental leave policies can shape respondents' attitudes toward the gendered division of parental leave in Germany. Other recent examples include topics such as how racial composition influences taxation preferences (O'Brien 2017), attitudes toward the denial of transgender

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healthcare rights (Doan & Grace 2022), the legitimacy of state intervention as a response to distant labor and environmental impacts (Amengual & Bartley 2022), and how correcting misinformation about local segregation impacts support for school desegregation policies (Thompson & Trejo 2024).

Second, survey experiments can be preferable when the outcome would be unethical or impossible to manipulate outside of the (hypothetical) survey context. For example, Tutić et al. (2022) ask respondents to make decisions about which patients should be offered a ventilator in a pandemic—clearly an unethical outcome were it not hypothetical! By developing the patient profiles using information from COVID-19 ventilator guidelines and policies, they can show when everyday people's decisions align with or deviate from the guidelines.

Other common examples include studies designed to test the causes of prejudice and bias. For example, Lu et al. (2021) use a survey experiment to demonstrate that priming Americans to think about the COVID-19 pandemic increases prejudice against Asian Americans. Similarly, Bonikowski & Zhang (2023) show how common messaging using populist and nationalist rhetoric can exacerbate exclusionary beliefs. In each case, intentionally promoting these beliefs in the real world could produce actual harm. Instead, the survey experiment allows the researchers to manipulate hypothesized factors driving negative attitudes in a hypothetical context (with debriefing materials as appropriate).[2]

Survey experiments are also useful when an outcome is not readily observable or easily measured, such as future plans or (unrealized) intentions. For instance, Pedulla & Thébaud (2015) show how institutional constraints and limited work-family policies offered by workplaces lead women to prefer more egalitarian arrangements in their future partnerships, whereas men are less responsive to institutional policy changes Similarly, Shepherd & Marshall (2018) examine how shared cultural representations can influence future fertility desires, and Robbins and colleagues (2022) examine norms around whether couples ought to marry. Because these future behaviors cannot be directly observed, a survey experiment is an appropriate research design.

A special case of outcomes not being readily observable is due to social desirability bias: A respondent might not be forthcoming with their true beliefs or behaviors, wanting to present themself in a positive light to the researcher (Levitt & List 2007). Survey experiments can be useful to combat social desirability bias because they can obscure the focus of the research.[3] For example, Flores & Schachter (2018) use a conjoint experiment that manipulates several treatments simultaneously to identify which factors lead an individual to be categorized as illegal (a stigmatized outcome). The design allows the authors to show that White Americans—including self-identified political liberals—tend to stereotype undocumented immigrants as criminals. Similarly, Sobotka & Stewart (2020) use a conjoint experiment to test which traits lead an individual to be categorized as someone who abuses opioids. In each case, the researchers manipulate multiple treatments simultaneously to give the respondents cover to express socially undesirable views (see Horiuchi et al. 2022).

Finally, survey experiments can be helpful for demonstrating how the measurement of a survey outcome itself influences findings. For example, survey experiments have demonstrated that the ways ethnoracial categories are measured—including how categorization questions are asked and the sets of categories offered as response options—shape whether people self-identify as Indigenous in Mexico (Flores et al. 2023), or as Middle Eastern and North African in the United States (Maghbouleh et al. 2022). Rivera & Tilcsik (2019) offer another example: In combination with a

[2]Institutional review board guidelines generally require debriefing materials in any research that involves deception, including the use of false information in experimental treatments.

[3]List experiments are another way to reduce social desirability bias (for more, see Blair & Imai 2012).

natural experiment, they use a survey experiment that varies the rating system in course evaluations from a 6-point scale to a 10-point scale to document how the scale properties themselves can affect the extent of gender bias in evaluations. By allowing researchers to causally manipulate the outcome format across a relatively large and diverse sample of respondents, survey experiments can be uniquely well-suited for these kinds of measurement studies (see also Schaeffer & Dykema 2020).

The Experimental Manipulations Are Impossible or Unethical Outside of a Hypothetical Context

Similar to outcomes, some experimental treatments cannot be ethically manipulated outside of a hypothetical context. For example, Mize & Manago (2018) use a vignette survey experiment to test the effects of same- and different-sex sexual encounters on the classification of individuals' sexual orientations. Randomly assigning participants to different sexual encounters would be unethical outside of a hypothetical context, necessitating the use of a survey experiment. They find that respondents are more likely to recategorize a heterosexual man's sexual orientation after a single same-sex sexual encounter than they are to question the sexual orientations of others. The findings underscore the precarity of heterosexual men's sexuality and help us understand why men holding this identity are so invested in protecting it. Similarly, Khanna & Meadow (2023) interrogate gender classification by manipulating a child's sex assigned at birth and their engagement in masculine/feminine-typed behavior. They find that people are more likely to reclassify boys exhibiting feminine behavior as transgender, compared with girls who exhibit masculine behavior, and offer a theoretical contribution to understanding the durability of maleness and femaleness. In another example, Almeling & Gadarian (2014) manipulate genetic risk scores for common diseases to measure how people respond to this health information. In each case, it could be unethical (if not impossible) to assign the treatment manipulations outside of a hypothetical context; survey experiments offer an ethical alternative.

The Counterfactuals Include Uncommon Combinations and/or Require an All-Else-Equal Comparison

Many sociological questions require a simultaneous focus on multiple characteristics that are co-occurring (Cho et al. 2013, Collins 2019, McCall 2005, Monk 2022). An experimental design offers one way for researchers to understand both the multiplicative and potentially independent effects of these characteristics, but doing so accurately often requires substantial statistical power—which, as discussed below, can be more accessible in the form of a survey experiment (Auspurg & Hinz 2015, Jasso 2006, Rossi 1979). For example, Schachter et al. (2021) use a conjoint survey experiment to test the relative effects of cultural cues, genetic ancestry, and skin color in shaping how White Americans racially categorize others. These traits are all distinct from one another but often co-occur so that it can be difficult to fully disentangle them using observational research methods. By independently randomizing each trait, they show that skin color has large effects on classification as White or Black and that skin color can override ancestry when the two signals conflict. These findings challenge the long-held assumption that the United States is primarily an ancestry-based racial system.

Survey experiments are also a useful tool when sociologists want to explicitly focus on the intersections of multiple traits. For example, Heiserman (2023) uses a factorial experimental design to assess intersectional stereotypes based on competence and warmth across gender, sexuality, race, age, and social class and finds a large amount of complexity, such that stereotypes associated with one social category often depend on one or more other categories (see also Pedulla 2014, Schachter 2021).

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Other times, sociologists use survey experiments to control for all explanations other than a key factor to offer an all-else-equal condition for a clear comparison. For example, many White Americans claim to prefer racially diverse neighborhoods but cite concerns about school quality and crime. Survey experiments allow urban sociologists to show that racial composition drives neighborhood preferences even after accounting for other neighborhood characteristics (Krysan et al. 2009). A similar approach has also been used to identify racial bias in school choice preferences (Billingham & Hunt 2016, Hailey 2022), racial bias in student discipline (Owens 2022), and bias based on sexual orientation (Beyerlein & Klocek 2020, Doan et al. 2015).

All-else-equal comparisons are also common approaches in studies of discrimination, and survey experiments have proven to be a beneficial methodological tool both as a standalone approach (e.g., Tilcsik 2021) and when combined with other methods to reveal the mechanisms of discrimination in decision-making (Pedulla 2016; Quadlin 2018; Rivera & Tilcsik 2016, 2023; Weisshaar 2018). In research on discrimination and the mechanisms of discrimination, social scientists are interested in whether a particular characteristic (such as gender, race, employment history, social class, or sexual orientation) causes unequal perceptions or differential treatment. To hone in on discrimination and the social-psychological mechanisms contributing to discriminatory behaviors, researchers need to isolate the characteristic(s) of interest and hold constant other factors that could have a (real or perceived) correlation with the characteristics of interest. For example, Weisshaar (2018) uses a population-based survey experiment to assess perceptions of job applicants that vary across their gender and employment history (employed, unemployed, and stay-at-home parent applicants who temporarily "opted out" of work), but have otherwise equivalent work experience and skills. The survey experiment illustrates that "opting out" to care for children signals a violation of ideal worker norms. Other studies of discrimination similarly vary only the characteristic of interest to isolate its relationship with discrimination (Pedulla 2016; Quadlin 2018; Rivera & Tilcsik 2016, 2023).

The Hypotheses Require Comparisons Across Multiple Subgroups and/or Large Sample Sizes for Statistical Power

Finally, because they can be embedded in surveys targeting theoretically relevant populations and/or subgroups, survey experiments can test hypotheses that require comparisons across multiple subgroups and/or large sample sizes for statistical power. While lab and field experiments can also be used for such hypotheses, the logistics and costs of participant recruitment can be prohibitive (Krupnikov & Findley 2016). As we explore in more detail below, survey experiments offer several distinct advantages, including relatively lower costs, the ability to capture extensive background information about participants, potential for repeated measurement via panel studies, and greater speed from design to implementation.[4]

Some studies build the subgroup comparisons into their original design. For example, Bloemraad et al. (2016) use a survey experiment to test how different framings influence support for immigrants' rights. They deliberately conduct the survey experiment on a large, diverse set of respondents that includes multiple political subgroups. This approach gives sufficient statistical power to test for subgroup differences, revealing important heterogeneity in how different frames can resonate with and generate support for immigrants' rights for one subgroup, even while alienating and generating opposition for another political subgroup. And Doan et al. (2014) use a vignette survey experiment to test how respondents' own gender and sexual orientation shape

[4]For more information on the potential logistical and financial strengths of survey experiments, readers are directed to **https://www.tessexperiments.org/info/introduction**.

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their willingness to offer formal rights (e.g., marriage) and informal privileges (e.g., displays of affection) to same-sex couples. They find that heterosexual respondents are willing to offer formal rights but less willing to extend informal privileges to same-sex couples, whereas lesbian and gay respondents distinguish less between these two types of rights.

Other studies use survey experiments to target subgroups to test or extend hypotheses identified in prior work. For example, Schachter (2016) uses a conjoint experiment to compare the effects of multiple characteristics, including race/ethnicity and legal status, on White Americans' perceptions of similarity with others. Adem & Ambriz (2023) offer an important addition, using a survey experiment to show how Black and Latinx Americans' conceptions of citizenship and belonging in the United States are different from those of Whites. And in the European context, scholars use a related experimental approach to study boundary-making in the context of rising Islamophobia (Krzyzowski & Nowicka 2021). Each of these types of studies offers causal evidence with respect to how particular groups of people respond differently to the same or similar treatments.

WHEN TO BE CAUTIOUS ABOUT USING SURVEY EXPERIMENTS

While survey experiments offer much value to sociology, they are not useful for all types of research questions. To assess the utility of a survey experiment as a method, it is helpful to first consider whether the goals of the study are suitable for an experiment at all and, if so, to then consider whether a survey experiment specifically is appropriate, or whether a lab or field experiment might be a preferable design (Jackson & Cox 2013).

First, are you asking a causal question informed by clear theories and predictions? A compelling experiment requires a solid experimental design (see more below), with clearly motivated experimental manipulations and theoretical predictions as to how the manipulation affects the outcomes. Experiments are excellent for hypothesis testing but are poorly suited for exploratory or inductive research, given that they require attention to the experimental manipulation and measurement before fielding the study. We recommend caution in using experiments for questions about multiple unrelated causes; exploratory work to narrow the study's focus is often warranted in these cases to avoid the number of experimental treatments becoming unwieldy.

Second, how well are the outcome of interest and the population of interest captured in a survey? As discussed above, outcomes related to categorization, attitudes, perceptions, and beliefs are well-suited for measurement in a survey, and coupled with a causal question, they may work well in a survey experiment. Outcomes that are more focused on behaviors can be appropriately assessed with survey experiment designs, but doing so often requires significant thought as to how the experiment will be designed to capture outcomes that accurately correspond to behaviors in real life (for example, hiring decisions, voting, policy utilization)—potentially by embedding a behavioral outcome in the survey context (e.g., Elliott-Negri et al. 2022, Finnigan & Hunter 2022, Tilcsik 2021). And, like in any survey, the ability to find participants that reflect the underlying population is critical (Marsden & Wright 2010) and may be especially important for certain experimental tasks that may only make sense for respondents with specific characteristics.

Finally, to what extent is the causal question translated into experimental manipulations in a survey context? If the research question is centrally focused on a social context, a field experiment might be a better fit (Baldassarri & Abascal 2017). Alternatively, if the research question is centrally focused on a process or interaction between two or more people, a lab experiment might be preferable (Bonacich & Light 1978, Jackson & Cox 2013). While survey respondents may be relatively easier to recruit, their accessibility is not a good reason to use a survey experiment

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BRIEF REFLECTIONS ON COMMON CRITIQUES OF SURVEY EXPERIMENTS

- *Survey experiments cannot produce realistic findings*: There are examples of survey experiments that produce findings close to real-life estimates of the same causal processes (e.g., Hainmueller et al. 2015). Moreover, the survey experimentalist's goal is often to establish internal validity for a causal test of a theoretical process that may not be observable in the real world (Mutz 2021).
- *Survey experiments are poor methods for studying behaviors*: Researchers should consider how well the behavioral measure can be understood by survey participants; how the stakes in the real world differ from the survey context; and whether a behavioral survey outcome, such as signing up for a newsletter or donating survey earnings, might be appropriate (see Elliott-Negri et al. 2022, Tilcsik 2021).
- *Survey experiments suffer from demand bias since participants can guess the experimental design*: Some survey experimental designs (conjoint experiments, between-subjects designs) can help reduce demand bias (Horiuchi et al. 2022, Levitt & List 2007).
- *Online survey participants cannot be trusted*: Online survey data collection options vary widely, but are not inherently untrustworthy. Researchers should follow best practices for avoiding bots or other low-quality responses (Alvarez et al. 2019, Coppock & McClellan 2019).

when a different type of experiment would offer more compelling evidence. For additional considerations on the suitability of survey experiments, see the sidebar titled Brief Reflections on Common Critiques of Survey Experiments.

BEST PRACTICES: SO YOU WANT TO DO A SURVEY EXPERIMENT?

Unlike more inductive approaches, in a survey experiment almost all consequential decisions happen prior to data collection. Investing in the design stage is essential as this is when the study is set up for either success or failure. Importantly, a successful survey experiment requires both thoughtful experimental design and survey design. In this section, we highlight some of the key issues and decisions researchers are likely to encounter, as well as practical steps to consider when moving from survey experiment design to implementation.

Experimental Design

There are a wide variety of designs that can work within a survey context; what is essential is random assignment. Random assignment is the critical tool that allows experiments to speak to causality by enabling a clean comparison across each experimental condition.

The design should be guided by the specific question and hypotheses. First, ask yourself: What comparisons do you need to make, given the theory and predictions you want to speak to? In other words, which experimental conditions will be necessary? For some types of questions, a traditional treatment and control or neutral condition might be suitable. For instance, Rivera et al. (2024) ask whether the inclusion of COVID-19 impact statements affects tenure evaluations; the control/neutral condition includes no impact statement, and the treatment conditions present one of two impact statements related to lab closures or childcare responsibilities.

Other questions might not use a control/neutral condition and instead involve comparisons across two or more treatment conditions. For example, in Gaddis's (2017) study testing the causal effects of first and last names to signal racial classification, there is no "deraced" name or traditional empty/neutral control condition. Rather, treatment effects are evaluated by comparing different randomly assigned name combinations to one another. The decision of whether to

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include a traditional control/neutral condition is therefore theory dependent (for more detailed discussions, see Jackson & Cox 2013, Mize & Manago 2022).

Asking which comparisons will be made can also inform the type of experimental design used. The comparisons could involve interaction effects across multiple treatments or could necessitate independent or additive comparisons across multiple groups. Some research questions lead to cumulative comparisons of the total effect of different scenarios or information on outcomes. No single design is better than another, but the design should be determined by which types of comparisons you want to make. We describe a few of the most common designs here.

Vignette experiments. Vignette designs can be an effective way to test whether different kinds of information frames influence attitudes or related outcomes. For example, Warren & Valentino (2023) use interviews to design vignettes to test Americans' understanding of omission as a form of racism. One potential challenge with vignette experiments is that if whole sentences, paragraphs, or sections of information differ across treatments, it is impossible to isolate which dimension(s) of the treatment are driving any effects. However, for research questions focused on the cumulative effect(s) of information, vignette experiments can be effective.[5]

Factorial experiments. In contrast to a vignette experiment, factorial experiments allow us to isolate independent or multiplicative effects of multiple attributes (Auspurg & Hinz 2015, Jasso 2006). By testing all possible attribute combinations, factorial experiments allow researchers to test for conditional effects. For example, Pedulla (2014) uses a factorial survey experiment design to test how holding multiple marginalized identities drives discrimination. His findings show that negative stereotypes about being gay hurt White men in the job application process but have positive consequences for Black men. It is only by comparing the treatment effects of all possible attribute combinations that this counterintuitive finding is revealed. Factorial designs can be employed in other forms of survey experiments, even within a vignette framework (Ishizuka 2019, Quadlin 2019). Importantly, in factorial experiments, the manipulations need to be fully crossed, such that each possible combination of attributes is presented, in order to test for conditional effects.

Conjoint experiments. One of the challenges of factorial experiments is that the required number of conditions to test all treatment combinations can grow rapidly, requiring high levels of statistical power generally only achieved via large sample sizes. In cases where scholars are more interested in the additive effects of multiple attributes rather than their interactive effects, conjoint designs are a viable alternative (Hainmueller et al. 2014).

Conjoint experiments differ from factorial designs in key ways. First, whereas factorial experiments are designed to test for interactions among treatments, conjoint experiments are designed to test for the independent, additive effects of multiple treatments.[6] Second, profiles (formatted as tables or vignettes) in conjoint designs are typically presented in pairs to increase respondent engagement, while in factorial designs, profiles/vignettes are typically presented one at a time. Third, it is standard to ask respondents to evaluate multiple profiles or pairs of profiles; this is an efficient way to generate sufficient statistical power. However, because respondents do not evaluate all possible profiles/treatment combinations, the multiple ratings per respondent are typically evaluated as independent observations in models that adjust for multiple observations from the same respondent using clustered standard errors.[7]

Vignette experiment: involves randomly assigning different sets of short text (vignettes) and comparing the cumulative effect of one vignette to another or to a control

Factorial experiment: involves manipulating two or more factors that are typically fully crossed, meaning all possible combinations of the manipulations are presented

Conjoint experiment: involves the manipulation of multiple experimental treatments; designed to test for independent, additive effects of multiple factors

[5]For more on best practices in vignette experiment design, readers are directed to Steiner et al. (2016).

[6]Importantly, we say "designed" here because both types of experiments can technically be used to perform both tests (i.e., identifying additive effects with a factorial design and interactions with a conjoint design) but are not optimized for both tests.

[7]For this reason, conjoint experiments are typically considered between-subjects designs even though respondents evaluate multiple profiles (see Hainmueller et al. 2014).

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Conjoint experiments became increasingly popular in the social sciences after political scientists Hainmueller et al. (2014) introduced a straightforward approach to estimating treatment effects (called average marginal component effects), but the design has its roots in sociology (see Jasso & Rossi 1977). While conjoint designs are becoming more popular in sociology (Adem & Ambriz 2023, Schachter 2016), research on best practices has continued to advance in other disciplines, especially political science (see De La Cuesta et al. 2022, Ganter 2023, Horiuchi et al. 2022, Leeper et al. 2020). Sociologists using conjoint experiments in their work must remain attuned to these ongoing developments happening outside of the discipline.

Within-subjects designs: the same respondent is randomly assigned to two or more experimental conditions and a treatment effect is calculated for each respondent

Between-subjects designs: randomization into treatment conditions across respondents is used for causal identification

Within- versus between-subject designs. Most survey experiments employ between-subjects designs where each participant is randomly assigned a single treatment, because they are easier to implement (Mutz 2021). While there are unique challenges in the design and application of within-subjects designs (Alferes 2013), they also have several strengths. First, because respondents act as their own control, within-subjects experiments require much smaller sample sizes relative to between-subjects designs (Clifford et al. 2021). Within-subjects designs also offer unique opportunities for causal mediation analysis (for more, see Acharya et al. 2018, Tomz & Weeks 2013), and we encourage sociologists to consider them.

Developing the Format and Wording of the Experimental Treatments and Survey Items

After finalizing the design, the next step is to develop the wording, formatting, and delivery of the treatments and the measurement of dependent variable(s), including question wording. While conceptualization should be theory-driven, how treatments and outcomes are represented and manipulated in the survey experiment should be informed by best practices in construct and survey design (Marsden & Wright 2010, Wolf et al. 2016).

Balancing realism and external validity with simplicity and internal validity in a survey context. The first—and sometimes correct—instinct might be to deliver the treatments in a form that is as close as possible to the way they are presented outside of a survey context. However, a survey almost never perfectly replicates the nonsurvey context to which scholars may want to make inferences. Decisions and preferences are often shaped by complex information and social context that may not translate to a survey, in which respondents' attention spans are short and nuanced details may be missed (Berinsky et al. 2014). Survey experimentalists therefore face a trade-off between designs that are complex but appear more realistic and those that are clear and easy-to-interpret tasks appropriate for surveys.

The essential task of a treatment is to cause variation in the outcome (dependent variable) (Mutz 2021). We urge researchers to design treatments with this goal foremost in mind, rather than prioritizing external validity above all else.[8] We are heartened by recent research suggesting the trade-offs between experimental control and external validity might not be as sharp as is often assumed (Brutger et al. 2023). Indeed, Hainmueller et al. (2015) exploit a natural experiment to demonstrate that survey experiment designs do not need to match actual choices made in the real world to make accurate inferences about behavior, though more research is needed in this area.

We recommend following best practices established in prior scholarship to guide decisions about treatments and wording. For example, there is extensive research documenting best

[8] If your primary goal is external validity, a survey experiment might not be your best choice for research design (see Gaines et al. 2007, Mutz 2021).

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practices for using names to manipulate race/ethnicity and social class (Gaddis 2017) or using photos and images to signal race/ethnicity (see Rich 2018).

Second, researchers should prioritize simplicity and clarity. No matter how complex or realistic the design, it will likely still be open to valid critiques about its external validity. Rather than prioritizing an elaborate design that most closely replicates actual choices or behaviors in the real world, focus on a design that clearly delivers your treatments and accurately measures your dependent variable. Once you establish an initial treatment effect, you can layer additional complexity into your design (see Sniderman 2018).

Survey items and dependent variable wording. Selecting the correct dependent variable is essential for a successful survey experiment. Developing clear language for the dependent variable (and all other survey questions) and their corresponding response categories is critical to a successful survey design; Schaeffer & Dykema (2020) provide a comprehensive review of best practices in question design. While a survey experiment only requires one dependent (outcome) variable, researchers often measure treatment effects on multiple outcomes—conceptualized as additional dependent variables or as potential mechanisms. Regardless, it is important to recognize that the first dependent variable measured immediately after the treatment(s) is administered is the only outcome that is fully causally identified (Gaines et al. 2007).

Pretesting: implementing all or part of a survey design with a convenience sample of respondents prior to the final version of the experiment

Power calculations: estimate necessary sample size to detect given effect size, or estimate effect size that could be observed for given sample size and statistical significance level

From Design to Implementation: Pretesting, Sample, and Preregistration

After developing the preliminary experimental design, there are several steps to take prior to finalizing and fielding the survey experiment. Here we highlight a few key aspects of this process, including pretesting, planning the survey sampling strategy, the role of preregistration, and best practices for the analysis of survey experiments.

Pretesting. Pretesting is an invaluable intermediate step but can also be abused or misused. Critically, pretesting allows researchers to test individual design components. For example, a pretest can measure how well different signals of music tastes are manipulating social class as intended (e.g., Thomas 2022). Here, the pretest outcome is not the experimental dependent variable, but a measure of perceived social class. This is a form of a manipulation check—testing the effectiveness of a treatment, which is sometimes retained in a final version of a survey experiment (Mutz 2021). Pretesting can also include open-ended questions to better gauge how respondents are understanding the dependent variable and/or overall experiment design. This kind of pretesting is especially important when researchers are using novel treatments that have not been validated by prior work.

Pretesting can also be used as a pilot study that evaluates the efficacy of the overall experiment. Here the idea is generally to use a smaller or less representative sample to establish treatment effects and demonstrate project viability prior to fielding the full study. Like pilot studies for other research designs, this type of pretesting can be especially helpful when researchers are seeking external funding, for example, to establish effect sizes for power calculations (Kenny & Judd 2019). Unfortunately, this form of pretesting has the potential to be abused if researchers keep iterating their experimental design until they finally find a statistically significant effect, and could contribute to part of the broader replication crisis in the social sciences (Dunning 2016, Freese & Peterson 2017, Munafò et al. 2017). We urge researchers to follow best practices in transparency and reproducibility during this type of pretesting.

Survey instrument and sample. Once the survey experiment design is finalized, researchers must consider the larger survey instrument. We recommend taking care to implement best practices for survey design (Krosnick 2018). In particular, we call for careful consideration of where within the

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flow of the survey the experimental manipulation or manipulations occur. Sometimes researchers need to collect pretreatment measures, but this runs the risk of influencing the experiment. And while the use of attention checks is common and can be helpful to identify low-quality respondents and/or bots (Alvarez et al. 2019), excluding these respondents can also lead to unbalanced assignment between treatment conditions (Kane et al. 2023).

In addition to paying careful attention to the entire survey instrument, scholars must also attend to the survey sample. The proliferation of low-cost online nonprobability survey samples has made survey experiments a more accessible and cost-effective method, though we encourage scholars to take care to maintain data quality (Chmielewski & Kucker 2020, Douglas et al. 2023, Mullinix et al. 2015, Peer et al. 2021). For questions focused on general social processes and/or detecting average treatment effects, these nonprobability opt-in samples may suffice (Coppock & McClellan 2019, Coppock et al. 2018). But as sociologists, often our questions are not about general or average effects—they are about how hiring managers evaluate job candidates (e.g., Pedulla 2016) or how members of different social groups navigate boundaries of race and citizenship (Adem & Ambriz 2023). These types of hypotheses often focus on particular groups of respondents and often include hypotheses about heterogenous treatment effects (e.g., that respondent characteristics moderate treatment effects). Following the framework proposed by Lundberg et al. (2021), we strongly encourage researchers to clearly identify their theoretical target population—the group for whom you hypothesize treatment effects—and any hypotheses about heterogeneity among the target population, and then select the survey sample that most closely matches your theorized target population.

Critically, for any given project budget, scholars will likely face trade-offs among the number of respondents, the representativeness of the sample (e.g., address-based, telephone-based, or opt-in), and how well the sample matches the theoretical target population (e.g., if the theoretical population is human resources managers, is the sample currently employed human resources managers, workers with hiring responsibilities, or something else?) (Marsden & Wright 2010). In general, as with other research methods, the smaller the population of interest, the more challenging it is to obtain a sample of respondents (Krupnikov & Findley 2016). And while the proliferation of online platforms has allowed for relatively low-cost and fast access to survey respondents, it has also led to increasing concerns about response quality (Chmielewski & Kucker 2020, Douglas et al. 2023, Mullinix et al. 2015, Peer et al. 2021).

When all three dimensions cannot be simultaneously optimized (as is often the case), we recommend navigating trade-offs based on the research question. Sample size contributes to statistical precision, which is an essential component of internal validity (Open Sci. Collab. 2015); underpowered studies risk imprecise identification of the causal effect of the treatment. Research aiming to establish a novel causal effect likely should prioritize sample size.

When the research question applies to a highly specific population, sample matching likely should be prioritized, especially if more representative sampling strategies are unavailable or prohibitively costly (Coppock et al. 2018, Krupnikov & Levine 2014, Mutz 2021). For research questions that apply to broader populations, obtaining a representative sample may be both more feasible and relatively more important to the research question. In practice researchers have used creative approaches to achieve a convincing sample for their research questions (for examples, see Owens 2022, Rivera et al. 2024).

Power analysis. Experiments are designed to test hypotheses about the effects of treatments. Critically, experiments must be designed to have sufficient statistical power to reject the null hypothesis of no treatment effect if the effect is, in fact, zero (Cohen 1988). Statistical power depends on the treatment effect size and the sample size; the smaller the true effect size, the greater the

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necessary sample size to detect the effect. Some funding agencies require power calculations to justify proposed sample sizes for experimental studies. In our experience, power analysis is most useful when researchers know (or have a good estimate of) the likely effect size of their treatment, which is not often the case for novel research. But even if precise estimates are not available, researchers must keep statistical power top of mind when designing their studies (for more on how to conduct power analyses, see Faul et al. 2007).

Preregistration: involves publishing one's analysis plan prior to data collection

Preregistration. Once the survey instrument is finalized, we encourage scholars to preregister their study by publicly sharing the analysis plan prior to data collection (see Nosek et al. 2018). Preregistration of experiments is common in other disciplines and encouraged or required by some sociology journals.[9] By making researchers commit to an experimental design and explicitly state their hypothesis test prior to data collection, preregistration aims to increase transparency and reduce post hoc fishing for statistically significant findings (Nosek et al. 2018). While preregistration might not entirely prevent practices like p-hacking (Franco et al. 2015, Jenke 2022), it is a useful tool for scholars to perform a final check that their experimental and survey designs map onto their hypotheses as well as possible. We recommend that sociologists conducting survey experiments preregister their studies prior to data collection.

Analysis and Other Considerations

While survey experiments generally involve relatively simple statistical analyses, we briefly highlight key analysis decisions researchers are likely to navigate. First, we have observed confusion as to whether survey weights should be used in the calculation of treatment effects (see Franco et al. 2017). Current best practice is to calculate and report both (unweighted) sample average treatment effects and (if available) population-level average treatment effects calculated using sampling weights. Any discrepancies point to issues with the sample weights, selection into the survey sample, and/or underlying heterogenous treatment effects (Miratrix et al. 2018; see also Huang 2024).

A second point of confusion is whether to include typical control variables in regressions when estimating treatment effects. If treatments are assigned randomly, the treatment assignment will be orthogonal to (uncorrelated with) respondent characteristics and including controls will not change the estimate of the treatment effect, though it can reduce standard errors (Berk et al. 2013). However, recent work suggests that the inclusion of covariates when estimating treatment effects (under conditions of clean random assignment) can introduce bias, and it is not recommended (see Mutz et al. 2019).

And third, sociologists often want to test hypotheses about interactions among treatments and/or between a treatment and a respondent characteristic. As discussed above, factorial experiments are explicitly designed to test interactions among treatments, and as long as the study is sufficiently powered, estimating heterogenous treatment effects by respondent characteristics is often as simple as estimating an interaction between treatment assignment and a respondent characteristic.

However, interaction effects are often underpowered. Taking advantage of advances in machine learning, scholars have developed new methods for estimating heterogenous treatment effects that address these limitations (see Athey & Imbens 2016, Grimmer et al. 2017, Ratkovic & Tingley 2023). Conjoint experiments offer additional challenges for estimating interactions both among

[9]Examples include the preregistration policies of the *American Journal of Sociology* (**https://www.journals.uchicago.edu/journals/ajs/further-guides-for-submission**) and *Sociological Science* (**https://sociologicalscience.com/reproducibility-policy/#Experiments**).

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BRIEF DISCUSSION OF COMMON PITFALLS WHEN CONDUCTING SURVEY EXPERIMENTS

- It is difficult to evaluate the soundness of a survey experiment without full information and transparency as to the design of the study and the intended analysis. We recommend preregistration and the use of supplemental/online appendices to inform readers about the design with sufficient detail, especially in cases when the researcher excludes certain participants, questions, or treatments from analysis.
- Comparisons across treatment conditions where multiple elements covary can limit the types of causal claims that can be made. Researchers should be careful to align their experimental design with the causal claims they want to make (and vice versa).
- Only the first question following a treatment is fully causally identified; researchers should avoid overstating claims of causality on other measures, including follow-up questions or open-ended responses. Using strategies like randomizing the order of questions can be helpful (Gaines et al. 2007).
- While large sample sizes can lead to "p-hacking" and other replicability concerns, underpowered comparisons can also be an issue—particularly when researchers claim that a lack of statistical significance indicates no causal effect (Cohen 1988).
- Researchers should prioritize simplicity and internal validity in their experimental designs; triangulate with other research methods to establish external validity (Mutz 2021, Gaines et al. 2007).

treatments (often termed causal interactions) and between treatments and respondent characteristics (often termed treatment heterogeneity). Recent methodological advances offer new tools for estimating interaction effects in conjoint experiments both among treatments (Egami & Imai 2019, Goplerud et al. 2024, Ham et al. 2024) and between treatments and respondent characteristics (Robinson & Duch 2024, Zhirkov 2022). For additional analytic considerations, please see the sidebar titled Brief Discussion of Common Pitfalls When Conducting Survey Experiments.

CONCLUDING THOUGHTS

We end this review by offering a few personal recommendations and observations regarding the future of survey experiments in sociology. First, we highlight recent innovative studies that reveal the strengths and limitations of survey experimental designs. We then describe the potential of mixed-method designs that pair survey experiments with other methods. Next, we explain why sociologists using survey experiments must pay careful attention to innovation in other disciplines. And finally, we call on the discipline to more actively participate in methodological conversation on survey experiments.

First, recent innovative studies have used survey experiments to gain leverage over how social, political, and/or historical contexts shape individuals' attitudes and behaviors. For example, Hart (2019) repeated the same survey experiment multiple times with repeated cross-sectional samples beginning with the start of the #MeToo movement to test whether the cultural bias against women who self-report sexual harassment at work would diminish as media attention to sexual harassment increased. Hart (2019) writes that, coincidentally, the first survey wave was launched on the same day that allegations against Harvey Weinstein first broke in the media; the addition of multiple survey waves to repeat the experiment over time was a novel methodological innovation that we encourage other scholars to follow. Another way sociologists can study change over time is through repeated surveying of the same respondents, a.k.a. a panel design. Several recent experiments have resurveyed the same participants multiple times to assess whether or how

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experimental stimuli produce lasting effects (Flores 2018) or to improve the realism of the task (Tilcsik 2021).

We also recommend that sociologists consider how a careful sampling strategy can allow for increased leverage over questions about context effects. Owens's (2022) study on racialized school discipline practices surveys teachers nested within a stratified probability sample of public middle and high schools to test how school context shapes teacher discipline practices. Other work considers the national-level context; for example, Mijs & Hoy (2022) field the same survey experiment exploring beliefs about meritocracy and inequality in three countries with varying levels of shared attitudes toward inequality among the population. Their cross-national comparative survey design allows them to show how national contexts shape responses to information about inequality, with key implications for future policy interventions. The proliferation of survey firms outside of the United States makes these kinds of research designs more accessible, and we encourage sociologists to take advantage.

Second, we see high potential for survey experiments to be included within larger mixed-methods projects, especially when paired with qualitative methods and/or computational social science methods. Qualitative methods in sociology, including interviews and ethnographic work, are generally not intended to establish causal relationships (Timmermans & Tavory 2012) but often offer rich, in-depth findings that are used in theory development. Survey experiments offer a unique opportunity to causally test newly developed theories, and, further, to consider variation in theoretical processes across contexts or respondents, in a way that complements qualitative findings. Few studies currently combine qualitative data analysis with a survey experiment (Rivera & Tilcsik 2016, Warren & Valentino 2023); in contrast, mixed-methods projects that combine survey experiments with field or natural experiments are relatively more common (e.g., Flores 2018, Weisshaar 2018). Some sociologists have recently bridged these methodological divides by including open-ended survey questions, for qualitative analysis, in their survey experiment designs (Grace & Doan 2024, Pepin & Scarborough 2018, Rivera et al. 2024), but we see potential for new and exciting developments in this vein, especially using computational approaches for text analysis.

For example, Thompson et al. (2023) use structural topic modeling (Grimmer et al. 2022) in their study of Black Americans' racial classification practices to estimate the causal effect of their survey experiment treatment conditions on respondents' open-ended text responses. The text analysis adds important nuance to their analysis, elucidating how Black Americans use different racial logics depending on the social context in which people are making claims to Blackness. We hope to see more sociologists employing this approach, which brings some of the richness of qualitative approaches to the causality and precision of a survey experiment.

Indeed, we see additional opportunities to pair survey experiments with computational methods, such as digital trace data (Macke et al. 2024). Work using machine learning and large-scale text corpuses has faced concerns about being able to detect causal relationships (Egami et al. 2022, Ermakoff 2019). Survey experiments, including those that use findings from computational analyses (Jackson & Brandtner 2024, Schachter et al. 2023), offer an avenue for assessing causal relationships between theoretical processes that may be otherwise difficult to estimate.

Looking beyond mixed-methods designs, we are also excited about ongoing research exploring how large language models like ChatGPT can augment social science research, including survey experiments (e.g., Ashokkumar et al. 2024). Another active area of development is adaptive arm experiments, which leverage algorithms to dynamically assign respondents to treatments to increase statistical efficiency (see Offer-Westort et al. 2021).

Survey experiments are a tool underutilized by sociologists (Mize & Manago 2022); we urge sociologists to consider integrating survey experiments into future projects. But doing so may

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hinge on how well sociology as a discipline can increase its expertise in and appreciation of the method. Indeed, while much of the historical development and innovation in survey experiment methodologies was performed by sociologists (Jasso & Rossi 1977), as of the writing of this review, most of this work is occurring in other disciplines, primarily political science and psychology [e.g., a 2022 special issue of *Political Analysis* (Jenke 2022)].

Sociologists using survey experiments in their research must be reading this work outside of the discipline. Some of the survey experiment designs used in our own prior work have since been shown to be suboptimal,[10] a development we view as core to scientific advancement. Even our recommendations within this review may soon be superseded by new advances. Sociologists using survey experiments must be part of this conversation.

Finally, as a multimethod discipline, we call on all sociologists, even those not using survey experiments themselves, to be a part of the conversation. Sociologists currently lack clear criteria for assessing the validity of survey experiments, including null findings, and lack publication outlets that are specifically geared toward experimental work. Since survey experiments involve multiple design and sampling decisions, it can be difficult for those without close familiarity with the method to evaluate the survey sample, question wording and response options, experimental treatments and controls, analysis, and results—and to weigh the importance of these factors. We hope that our review article begins to fill this void but recognize that more work is needed.

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LITERATURE CITED

Acharya A, Blackwell M, Sen M. 2018. Analyzing causal mechanisms in survey experiments. *Political Anal.* 26(4):357–78

Adem M, Ambriz D. 2023. What makes a citizen? Contemporary immigration and the boundaries of citizenry. *Soc. Forces* 102(2):730–52

Alferes VR. 2013. Within-subjects designs randomization. In *Methods of Randomization in Experimental Design*, ed. VR Alferes, pp. 65–106. Thousand Oaks, CA: SAGE

Almeling R, Gadarian SK. 2014. Reacting to genetic risk: an experimental survey of life between health and disease. *J. Health Soc. Behav.* 55(4):482–503

Alvarez RM, Atkeson LR, Levin I, Li Y. 2019. Paying attention to inattentive survey respondents. *Political Anal.* 27(2):145–62

Amengual M, Bartley T. 2022. Global markets, corporate assurances, and the legitimacy of state intervention: perceptions of distant labor and environmental problems. *Am. Sociol. Rev.* 87(3):383–414

Ashokkumar A, Hewitt L, Ghezae I, Willer R. 2024. *Predicting results of social science experiments using large language models*. Work. Pap., New York Univ., New York, NY

[10]For example, the experiment in Schachter (2016) includes randomization restrictions among the treatments. While technically possible, restrictions in conjoint experiments limit the interpretation of the results and today are not recommended. This experimental design also uses a forced choice between two randomly assigned profiles. Methodological advances published after the survey in Schachter (2016) was fielded now recommend caution in using forced-choice designs (see Hainmueller et al. 2015, Ganter 2023).

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Athey S, Imbens G. 2016. Recursive partitioning for heterogeneous causal effects. *PNAS* 113(27):7353–60
Auspurg K, Hinz T. 2015. *Factorial Survey Experiments*. Thousand Oaks, CA: SAGE
Baldassarri D, Abascal M. 2017. Field experiments across the social sciences. *Annu. Rev. Sociol.* 43:41–73
Berinsky AJ, Margolis MF, Sances MW. 2014. Separating the shirkers from the workers? Making sure respondents pay attention on self-administered surveys. *Am. J. Political Sci.* 58(3):739–53
Berk R, Pitkin E, Brown L, Buja A, George E, Zhao L. 2013. Covariance adjustments for the analysis of randomized field experiments. *Eval. Rev.* 37(3–4):170–96
Beyerlein K, Klocek J. 2020. How do religion and sexual orientation affect support for U.S. presidential candidates? Evidence from a survey experiment. *J. Sci. Study Relig.* 59(4):551–68
Billingham CM, Hunt MO. 2016. School racial composition and parental choice: new evidence on the preferences of White parents in the United States. *Sociol. Educ.* 89(2):99–117
Blair G, Imai K. 2012. Statistical analysis of list experiments. *Political Anal.* 20(1):47–77
Bloemraad I, Silva F, Voss K. 2016. Rights, economics, or family? Frame resonance, political ideology, and the immigrant rights movement. *Soc. Forces* 94(4):1647–74
Bonacich P, Light J. 1978. Laboratory experimentation in sociology. *Annu. Rev. Sociol.* 4:145–70
Bonikowski B, Zhang Y. 2023. Populism as dog-whistle politics: anti-elite discourse and sentiments toward minority groups. *Soc. Forces* 102(1):180–201
Brutger R, Kertzer JD, Renshon J, Tingley D, Weiss CM. 2023. Abstraction and detail in experimental design. *Am. J. Political Sci.* 67(4):979–95
Chmielewski M, Kucker SC. 2020. An MTurk crisis? Shifts in data quality and the impact on study results. *Soc. Psychol. Personal. Sci.* 11(4):464–73
Cho YI, Johnson TP, VanGeest JB. 2013. Enhancing surveys of health care professionals: a meta-analysis of techniques to improve response. *Eval. Health Prof.* 36(3):382–407
Clifford S, Sheagley G, Piston S. 2021. Increasing precision without altering treatment effects: repeated measures designs in survey experiments. *Am. Political Sci. Rev.* 115(3):1048–65
Cohen J. 1988. *Statistical Power Analysis for the Behavioral Sciences*. New York: Routledge. 2nd ed.
Collins PH. 2019. *Intersectionality as Critical Social Theory*. Durham, NC: Duke Univ. Press
Coppock A, Leeper TJ, Mullinix KJ. 2018. Generalizability of heterogeneous treatment effect estimates across samples. *PNAS* 115(49):12441–46
Coppock A, McClellan OA. 2019. Validating the demographic, political, psychological, and experimental results obtained from a new source of online survey respondents. *Res. Politics* 6(1). **https://doi.org/10.1177/2053168018822174**
De La Cuesta B, Egami N, Imai K. 2022. Improving the external validity of conjoint analysis: the essential role of profile distribution. *Political Anal.* 30(1):19–45
Doan L, Grace MK. 2022. Factors affecting public opinion on the denial of healthcare to transgender persons. *Am. Sociol. Rev.* 87(2):275–302
Doan L, Loehr A, Miller LR. 2014. Formal rights and informal privileges for same-sex couples: evidence from a national survey experiment. *Am. Sociol. Rev.* 79(6):1172–95
Doan L, Miller LR, Loehr A. 2015. The power of love: the role of emotional attributions and standards in heterosexuals' attitudes toward lesbian and gay couples. *Soc. Forces* 94(1):401–25
Douglas BD, Ewell PJ, Brauer M. 2023. Data quality in online human-subjects research: comparisons between MTurk, Prolific, CloudResearch, Qualtrics, and SONA. *PLOS ONE* 18(3):e0279720
Druckman JN. 2022. *Experimental Thinking: A Primer on Social Science Experiments*. Cambridge, UK: Cambridge Univ. Press. 1st ed.
Druckman JN, Green DP, Kuklinski JH, Lupia A, eds. 2011. *Cambridge Handbook of Experimental Political Science*. Cambridge, UK: Cambridge Univ. Press
Dunning T. 2012. *Natural Experiments in the Social Sciences: A Design-Based Approach*. Cambridge, UK: Cambridge Univ. Press
Dunning T. 2016. Transparency, replication, and cumulative learning: what experiments alone cannot achieve. *Annu. Rev. Political Sci.* 19:541–63
Egami N, Fong CJ, Grimmer J, Roberts ME, Stewart BM. 2022. How to make causal inferences using texts. *Sci. Adv.* 8(42):eabg2652

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Egami N, Imai K. 2019. Causal interaction in factorial experiments: application to conjoint analysis. *J. Am. Stat. Assoc.* 114(526):529–40

Elliott-Negri L, Tu S, Zheng W, Lennon MC. 2022. Hope, emotional charges, and online action: an experimental study of the DREAM Act. *Soc. Probl.* 69(3):781–98

Elwert F, Winship C. 2014. Endogenous selection bias: the problem of conditioning on a collider variable. *Annu. Rev. Sociol.* 40:31–53

Ermakoff I. 2019. Causality and history: modes of causal investigation in historical social sciences. *Annu. Rev. Sociol.* 45:581–606

Faul F, Erdfelder E, Lang A-G, Buchner A. 2007. G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behav. Res. Methods* 39(2):175–91

Finnigan R, Hunter S. 2022. Policy regulation of precarious work schedules and bottom-up enforcement: an evaluation of state reporting pay policies. *Soc. Forces* 101(2):747–73

Flores RD. 2018. Can elites shape public attitudes toward immigrants? Evidence from the 2016 US presidential election. *Soc. Forces* 96(4):1649–90

Flores RD, Schachter A. 2018. Who are the "illegals"? The social construction of illegality in the United States. *Am. Sociol. Rev.* 83(5):839–68

Flores RD, Vignau-Loría M, Martínez Casas R. 2023. Transitory versus durable boundary crossing: What explains the Indigenous population boom in Mexico? *Am. J. Sociol.* 129(1):123–61

Franco A, Malhotra N, Simonovits G. 2015. Underreporting in political science survey experiments: comparing questionnaires to published results. *Political Anal.* 23(2):306–12

Franco A, Malhotra N, Simonovits G, Zigerell LJ. 2017. Developing standards for post-hoc weighting in population-based survey experiments. *J. Exp. Political Sci.* 4(2):161–72

Freese J, Peterson D. 2017. Replication in social science. *Annu. Rev. Sociol.* 43:147–65

Gaddis S. 2017. How Black are Lakisha and Jamal? Racial perceptions from names used in correspondence audit studies. *Sociol. Sci.* 4:469–89

Gaines BJ, Kuklinski JH, Quirk PJ. 2007. The logic of the survey experiment reexamined. *Political Anal.* 15(1):1–20

Ganter F. 2023. Identification of preferences in forced-choice conjoint experiments: reassessing the quantity of interest. *Political Anal.* 31(1):98–112

Goplerud M, Imai K, Pashley NE. 2024. Estimating heterogeneous causal effects of high-dimensional treatments: application to conjoint analysis. arXiv:2201.01357v4 [stat.ME]

Grace MK, Doan L. 2024. Medical authority, trans exceptionalism, and Americans' willingness to believe claims of inadequate training as justification for the denial of care to trans people. *Socius* 10. **https://doi.org/10.1177/23780231241253969**

Grimmer J, Messing S, Westwood SJ. 2017. Estimating heterogeneous treatment effects and the effects of heterogeneous treatments with ensemble methods. *Political Anal.* 25(4):413–34

Grimmer J, Roberts ME, Stewart BM. 2022. *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton, NJ: Princeton Univ. Press

Hailey CA. 2022. Racial preferences for schools: evidence from an experiment with White, Black, Latinx, and Asian parents and students. *Sociol. Educ.* 95(2):110–32

Hainmueller J, Hangartner D, Yamamoto T. 2015. Validating vignette and conjoint survey experiments against real-world behavior. *PNAS* 112(8):2395–400

Hainmueller J, Hopkins DJ, Yamamoto T. 2014. Causal inference in conjoint analysis: understanding multidimensional choices via stated preference experiments. *Political Anal.* 22(1):1–30

Ham DW, Imai K, Janson L. 2024. Using machine learning to test causal hypotheses in conjoint analysis. *Political Anal.* 32:329–44

Hart CG. 2019. The penalties for self-reporting sexual harassment. *Gend. Soc.* 33(4):534–59

Heiserman N. 2023. Intersectional complexity in stereotype content. *Am. J. Sociol.* 129(3):637–1032

Holland PW. 1986. Statistics and causal inference. *J. Am. Stat. Assoc.* 81(396):945–60

Horiuchi Y, Markovich Z, Yamamoto T. 2022. Does conjoint analysis mitigate social desirability bias? *Political Anal.* 30(4):535–49

Huang MY. 2024. Sensitivity analysis for the generalization of experimental results. *J. R. Stat. Soc. Ser. A* 187(4):900–18

Ishizuka P. 2019. Social class, gender, and contemporary parenting standards in the United States: evidence from a national survey experiment. *Soc. Forces* 98(1):31–58

Jackson M, Brandtner C. 2024. Institutional exclusion: the cultural production of educational inequality through college narratives. SocArXiv. **https://doi.org/10.31235/osf.io/3yfjz**

Jackson M, Cox DR. 2013. The principles of experimental design and their application in sociology. *Annu. Rev. Sociol.* 39:27–49

Jasso G. 2006. Factorial survey methods for studying beliefs and judgments. *Sociol. Methods Res.* 34(3):334–423

Jasso G, Rossi PH. 1977. Distributive justice and earned income. *Am. Sociol. Rev.* 42:639–51

Jenke L. 2022. Introduction to the special issue: innovations and current challenges in experimental methods. *Political Anal.* 30(S1):S3–7

Kane JV, Velez YR, Barabas J. 2023. Analyze the attentive and bypass bias: mock vignette checks in survey experiments. *Political Sci. Res. Methods* 11(2):293–310

Kenny DA, Judd CM. 2019. The unappreciated heterogeneity of effect sizes: implications for power, precision, planning of research, and replication. *Psychol. Methods* 24(5):578–89

Khanna K, Meadow T. 2023. The fragile male: an experimental study of transgender classification and the durability of gender categories. *Gend. Soc.* 37(4):553–83

Krosnick JA. 2018. Questionnaire design. In *The Palgrave Handbook of Survey Research*, ed. DL Vannette, JA Krosnick, pp. 439–55. Cham, Switz.: Springer

Krupnikov Y, Findley B. 2016. Survey experiments: managing the methodological costs and benefits. In *The Oxford Handbook of Polling and Survey Methods*, Vol. 1, ed. LR Atkeson, RM Alvarez, pp. 483–504. Oxford, UK: Oxford Univ. Press

Krupnikov Y, Levine AS. 2014. Cross-sample comparisons and external validity. *J. Exp. Political Sci.* 1(1):59–80

Krysan M, Couper MP, Farley R, Forman TA. 2009. Does race matter in neighborhood preferences? Results from a video experiment. *Am. J. Sociol.* 115(2):527–59

Krzyzowski L, Nowicka M. 2021. European solidarity as boundary-making: a conjoint analysis of attitudes towards Islam in the context of the 'refugee crisis.' *J. Sociol.* 57(2):305–24

Leeper TJ, Hobolt SB, Tilley J. 2020. Measuring subgroup preferences in conjoint experiments. *Political Anal.* 28(2):207–21

Levitt SD, List JA. 2007. What do laboratory experiments measuring social preferences reveal about the real world? *J. Econ. Perspect.* 21(2):153–74

Lu Y, Kaushal N, Huang X, Gaddis SM. 2021. Priming COVID-19 salience increases prejudice and discriminatory intent against Asians and Hispanics. *PNAS* 118(36):e2105125118

Lundberg I, Johnson R, Stewart BM. 2021. What is your estimand? Defining the target quantity connects statistical evidence to theory. *Am. Sociol. Rev.* 86(3):532–65

Macke E, Daviss C, Williams-Baron E. 2024. Untapped potential: collecting and analyzing digital trace data within survey experiments. *Sociol. Methods Res.* In press. **https://doi.org/10.1177/00491241241268770**

Maghbouleh N, Schachter A, Flores RD. 2022. Middle Eastern and North African Americans may not be perceived, nor perceive themselves, to be White. *PNAS* 119(7):e2117940119

Marsden PV, Wright JD. 2010. *Handbook of Survey Research*. Bingley, UK: Emerald

McCall L. 2005. The complexity of intersectionality. *Signs J. Women Cult. Soc.* 30(3):1771–800

Mijs JJB, Hoy C. 2022. How information about inequality impacts belief in meritocracy: evidence from a randomized survey experiment in Australia, Indonesia and Mexico. *Soc. Probl.* 69(1):91–122

Miratrix LW, Sekhon JS, Theodoridis AG, Campos LF. 2018. Worth weighting? How to think about and use weights in survey experiments. *Political Anal.* 26(3):275–91

Mize TD, Manago B. 2018. Precarious sexuality: how men and women are differentially categorized for similar sexual behavior. *Am. Sociol. Rev.* 83(2):305–30

Mize TD, Manago B. 2022. The past, present, and future of experimental methods in the social sciences. *Soc. Sci. Res.* 108:102799

Monk EP. 2022. Inequality without groups: contemporary theories of categories, intersectional typicality, and the disaggregation of difference. *Sociol. Theory* 40(1):3–27

Mullinix KJ, Leeper TJ, Druckman JN, Freese J. 2015. The generalizability of survey experiments. *J. Exp. Political Sci.* 2(2):109–38

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Munafò MR, Nosek BA, Bishop DVM, Button KS, Chambers CD, et al. 2017. A manifesto for reproducible science. *Nat. Hum. Behav.* 1:21

Mutz DC. 2021. Improving experimental treatments in political science. In *Advances in Experimental Political Science*, ed. JN Druckman, DP Green, pp. 219–38. Cambridge, UK: Cambridge Univ. Press

Mutz DC, Pemantle R, Pham P. 2019. The perils of balance testing in experimental design: messy analyses of clean data. *Am. Stat.* 73(1):32–42

Nosek BA, Ebersole CR, DeHaven AC, Mellor DT. 2018. The preregistration revolution. *PNAS* 115(11):2600–6

O'Brien RL. 2017. Redistribution and the new fiscal sociology: race and the progressivity of state and local taxes. *Am. J. Sociol.* 122(4):1015–49

Offer-Westort M, Coppock A, Green DP. 2021. Adaptive experimental design: prospects and applications in political science. *Am. J. Political Sci.* 65(4):826–44

Open Sci. Collab. 2015. Estimating the reproducibility of psychological science. *Science* 349(6251):aac4716

Owens J. 2022. Double jeopardy: teacher biases, racialized organizations, and the production of racial/ethnic disparities in school discipline. *Am. Sociol. Rev.* 87(6):1007–48

Pedulla DS. 2014. The positive consequences of negative stereotypes: race, sexual orientation, and the job application process. *Soc. Psychol. Q.* 77(1):75–94

Pedulla DS. 2016. Penalized or protected? Gender and the consequences of nonstandard and mismatched employment histories. *Am. Sociol. Rev.* 81(2):262–89

Pedulla DS, Thébaud S. 2015. Can we finish the revolution? Gender, work-family ideals, and institutional constraint. *Am. Sociol. Rev.* 80(1):116–39

Peer E, Rothschild D, Gordon A, Evernden Z, Damer E. 2021. Data quality of platforms and panels for online behavioral research. *Behav. Res. Methods* 54(4):1643–62

Pepin JR, Scarborough WJ. 2018. What makes a decision fair? Relative earnings, gender, and justifications for couples' decision making. SocArXiv. **https://doi.org/10.31235/osf.io/4dynh**

Philipp M-F, Büchau S, Schober PS, Spiess CK. 2023. Parental leave policies, usage consequences, and changing normative beliefs: evidence from a survey experiment. *Gend. Soc.* 37(4):493–523

Quadlin N. 2018. The mark of a woman's record: gender and academic performance in hiring. *Am. Sociol. Rev.* 83(2):331–60

Quadlin N. 2019. Sibling achievement, sibling gender, and beliefs about parental investment: evidence from a national survey experiment. *Soc. Forces* 97(4):1603–30

Ratkovic M, Tingley D. 2023. Estimation and inference on nonlinear and heterogeneous effects. *J. Politics* 85(2):421–35

Rich J. 2018. Do photos help or hinder field experiments of discrimination? *Int. J. Manpow.* 39(4):502–18

Rivera LA, Tilcsik A. 2016. Class advantage, commitment penalty: the gendered effect of social class signals in an elite labor market. *Am. Sociol. Rev.* 81(6):1097–131

Rivera LA, Tilcsik A. 2019. Scaling down inequality: rating scales, gender bias, and the architecture of evaluation. *Am. Sociol. Rev.* 84(2):248–74

Rivera LA, Tilcsik A. 2023. Not in my schoolyard: disability discrimination in educational access. *Am. Sociol. Rev.* 88(2):284–321

Rivera LA, Weisshaar K, Tilcsik A. 2024. Disparate impact? Career disruptions and COVID-19 impact statements in tenure evaluations. *Sociol. Sci.* 11:626–48

Robbins BG, Dechter A, Kornrich S. 2022. Assessing the deinstitutionalization of marriage thesis: an experimental test. *Am. Sociol. Rev.* 87(2):237–74

Robinson TS, Duch RM. 2024. How to detect heterogeneity in conjoint experiments. *J. Politics* 86(2):412–27

Rossi PH. 1979. Vignette analysis: uncovering the normative structure of complex judgments. In *Qualitative and Quantitative Social Research: Papers in Honor of Paul F. Lazarsfeld*, ed. RK Merton, JS Coleman, PH Rossi, pp. 176–86. New York: Free Press

Schachter A. 2016. From "different" to "similar": an experimental approach to understanding assimilation. *Am. Sociol. Rev.* 81(5):981–1013

Schachter A. 2021. Intersecting boundaries: comparing stereotypes of native- and foreign-born members of ethnoracial groups. *Soc. Forces* 100(2):506–39

Downloaded from www.annualreviews.org. Guest (guest) IP: 201.39.147.100 On: Fri, 16 Oct 2026 13:27:26

Schachter A, Flores RD, Maghbouleh N. 2021. Ancestry, color, or culture? How Whites racially classify others in the U.S. *Am. J. Sociol.* 126(5):1220–63
Schachter A, Kuk J, Besbris M, Pekarek G. 2023. Inclusive but not integrative: ethnoracial boundaries and the use of Spanish in the market for rental housing. *Sociol. Sci.* 10:585–612
Schaeffer NC, Dykema J. 2020. Advances in the science of asking questions. *Annu. Rev. Sociol.* 46:37–60
Shepherd H, Marshall EA. 2018. The implicit activation mechanism of culture: a survey experiment on associations with childbearing. *Poetics* 69:1–14
Sniderman PM. 2018. Some advances in the design of survey experiments. *Annu. Rev. Political Sci.* 21:259–75
Sobotka TC, Stewart SA. 2020. Stereotyping and the opioid epidemic: a conjoint analysis. *Soc. Sci. Med.* 255:113018
Steiner PM, Atzmüller C, Su D. 2016. Designing valid and reliable vignette experiments for survey research: a case study on the fair gender income gap. *J. Methods Meas. Soc. Sci.* 7(2):52–94
Thomas K. 2022. The psychology of distinction: how cultural tastes shape perceptions of class and competence in the U.S. *Poetics* 93:101669
Thompson ME, Trejo S. 2024. My school district isn't segregated: experimental evidence on the effect of information on parental preferences regarding school segregation. *Sociol. Educ.* 97(2):97–119
Thompson ME, Trejo S, Alvero A, Martschenko D. 2023. "They have Black in their blood": exploring how genetic ancestry tests affect racial appraisals and classifications. SocArXiv. **https://doi.org/10.31235/osf.io/8tnrk**
Tilcsik A. 2021. Statistical discrimination and the rationalization of stereotypes. *Am. Sociol. Rev.* 86(1):93–122
Timmermans S, Tavory I. 2012. Theory construction in qualitative research: from grounded theory to abductive analysis. *Sociol. Theory* 30(3):167–86
Tomz MR, Weeks JLP. 2013. Public opinion and the democratic peace. *Am. Political Sci. Rev.* 107(4):849–65
Tutić A, Krumpal I, Haiser F. 2022. Triage in times of COVID-19: a moral dilemma. *J. Health Soc. Behav.* 63(4):560–76
Warren E, Valentino L. 2023. Inaction, silence, focus, and power: identifying and assessing folk theories of the racism of omission. *Soc. Probl.* 2023:spad054
Weisshaar K. 2018. From opt out to blocked out: the challenges for labor market re-entry after family-related employment lapses. *Am. Sociol. Rev.* 83(1):34–60
Wolf C, Joye D, Smith TW, Fu Y, eds. 2016. *The Sage Handbook of Survey Methodology*. Thousand Oaks, CA: SAGE
Zhirkov K. 2022. Estimating and using individual marginal component effects from conjoint experiments. *Political Anal.* 30(2):236–49

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